

Binary Digits

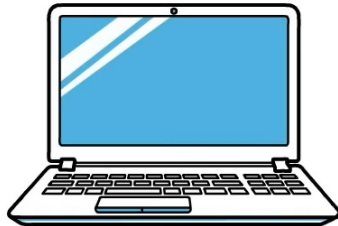
Bi → Two

0

👉 Low value

👉 Off

👉 False



Use binary digits

1

👉 High value

👉 On

👉 True

Boolean Algebra

We use binary digits.

$$1 + 1 = 1$$

Addition

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 1$$

Multiplication

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

$$1 \cdot 1 = 1$$

Zero x 9999 = Hydrogen bomb

Concepts of Boolean Algebra

Addition of Binary digits

$A = 0$

$A = 1$



$$A + 1 = 1$$

$$0 + 1 = 1$$

$$1 + 1 = 1$$



$$A + 0 = A$$

$$0 + 0 = 0$$

$$1 + 0 = 1$$

Multiplication of Binary digits

$A = 0$

$A = 1$



$$A . 1 = A$$

$$0 . 1 = 0$$

$$1 . 1 = 1$$



$$A . 0 = 0$$

$$0 . 0 = 0$$

$$1 . 0 = 0$$

Inverse concept

Low value

High value

0 $\longrightarrow$ 1

A $\longrightarrow$ $\bar{A}$



$$A + \bar{A} = 1$$

$A = 0$ $\bar{A} = 1$

$$0 + 1 = 1$$



$$\bar{\bar{A}} = A$$

$A = 1$ $\bar{A} = 0$ $\bar{\bar{A}} = 1$



$$A . \bar{A} = 0$$

$$0 . 1 = 0$$

Advance Concept of Boolean Algebra

Proof

$$\overline{(A+B)} = \bar{A} \cdot \bar{B}$$

$$\overline{(A \cdot B)} = \bar{A} + \bar{B}$$

LHS: $\overline{(A+B)}$

$A=1 \quad B=0$

👉 $(A+B)$

$= 1+0$

$= 1$

👉 $\overline{(A+B)}$

$= 1$

$\boxed{= 0} \dots \text{eq(1)}$

RHS: $\bar{A} \cdot \bar{B}$

$A=1 \quad B=0 \quad \bar{A}=0 \quad \bar{B}=1$

👉 $\bar{A} \cdot \bar{B}$

$= 0 \cdot 1$

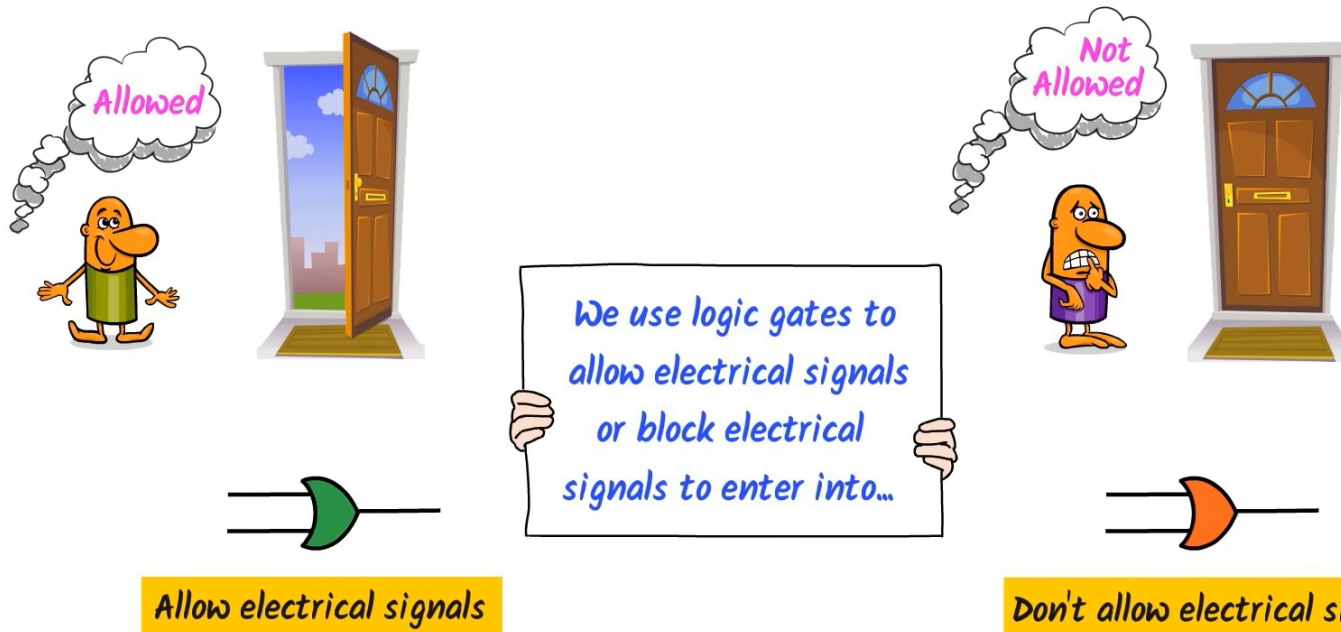
$\boxed{= 0} \dots \text{eq(2)}$

$\text{eq(1)} = \text{eq(2)}$

$\text{LHS} = \text{RHS}$

Concept of Logic Gates

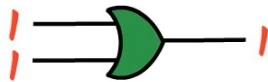
Purpose of Gates



What are Logic Gates ?

Two Inputs $\Rightarrow$ One Output

"The digital devices which perform Boolean function by taking two inputs and giving one output."



Allow electrical signals



Logic gates are based on Boolean Algebra.



Logic gates are made up of diode and transistor.

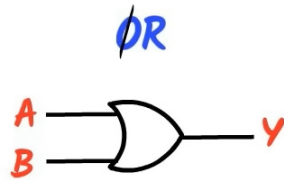


Don't allow electrical signals



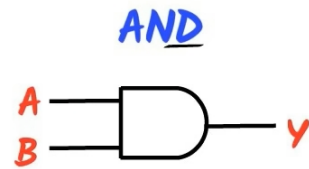
Types of Logic Gates

5 Universal Logic Gates



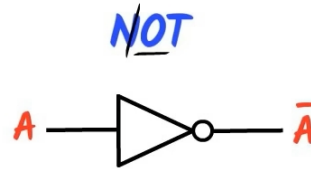
$$Y = A + B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1



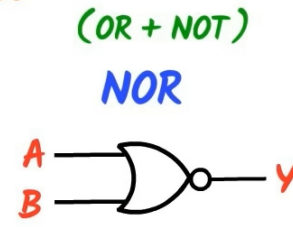
$$Y = A \cdot B$$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



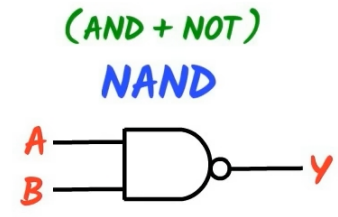
$$Y = \bar{A}$$

A	Y
0	1
1	0



$$Y = \overline{A + B}$$

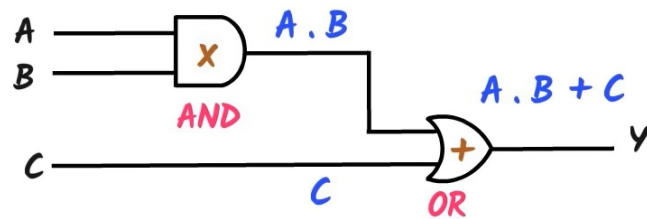
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0



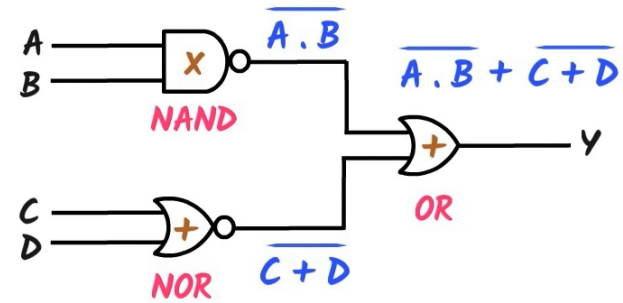
$$Y = \overline{A \cdot B}$$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Writing Functions for Logic Gates



$$Y = (A.B) + C$$



$$Y = (\overline{A.B}) + (\overline{C+D})$$

Exam Questions

What will be the value of A and B for the Boolean expression $\overline{(A+B)} \cdot (\bar{A} \cdot \bar{B}) = 1$?

$$A = 0 \quad B = 0$$

$$\text{LHS: } \overline{(A+B)} \cdot (\bar{A} \cdot \bar{B})$$

$$= \overline{(0+0)} \cdot (\bar{0} \cdot \bar{0})$$

$$= (\bar{0}) \cdot 1 \cdot 1$$

$$= 1 \cdot 1 \cdot 1$$

$$= 1$$

Exam Questions

👁 Show

This logic gate is equivalent to OR

